

Reading 1

Agile Fundamental and Scrum Framework



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Topic 1

What is Agile and Why it matters

Outline



Introduction to
VUCA

the Agile Manifesto

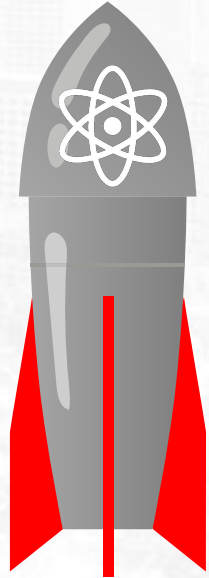
4 Agile Values

12 Agile Principles

Introduction to VUCA

History behind VUCA

- First used in 1987
- Based on leadership theories of Warren Bennis and Burt Nanus
- As response of the collapse of the USSR by the end of cold war





Volatility
Uncertainty
Complexity
Ambiguity

Volatility

Behavior

- Unstable and constantly changing
- Not necessarily hard to understand
- Knowledge often available

Response

- Prepare, reserve, threshold



Uncertainty

Behavior

- Unexpected events
- Difficult to anticipate or predict
- Historical forecasts & past experiences losing relevance

Response

- Collect information, interpret & share



Complexity

Behavior

- Multiple changes and events happened simultaneously
- Certain cause trigger interconnected multi layered reaction, parts and variables
- Overwhelming to process

Response

- Inspect & Adapt



Ambiguity

Behavior

- Right or wrong completely unclear
- Whole new unknown environment & definition

Response

- Experimenting





VUCA simply happen around us
Most likely we are unable to change the situation
What we can do is to adapt our behavior accordingly

Further Readings

- <https://hbr.org/2014/01/what-vuca-really-means-for-you>
- <https://www.vuca-world.org/>

Outline

1

**Introduction to
VUCA**



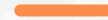
2

the Agile Manifesto



3

4 Agile Values



4

12 Agile Principles

The Agile Manifesto

History Behind The Agile Manifesto

- 17 software development practitioner gather at a ski resort in 2001
- Shares common obstacles in delivering valuable software
- First called as lightweight process / framework for more effective software development



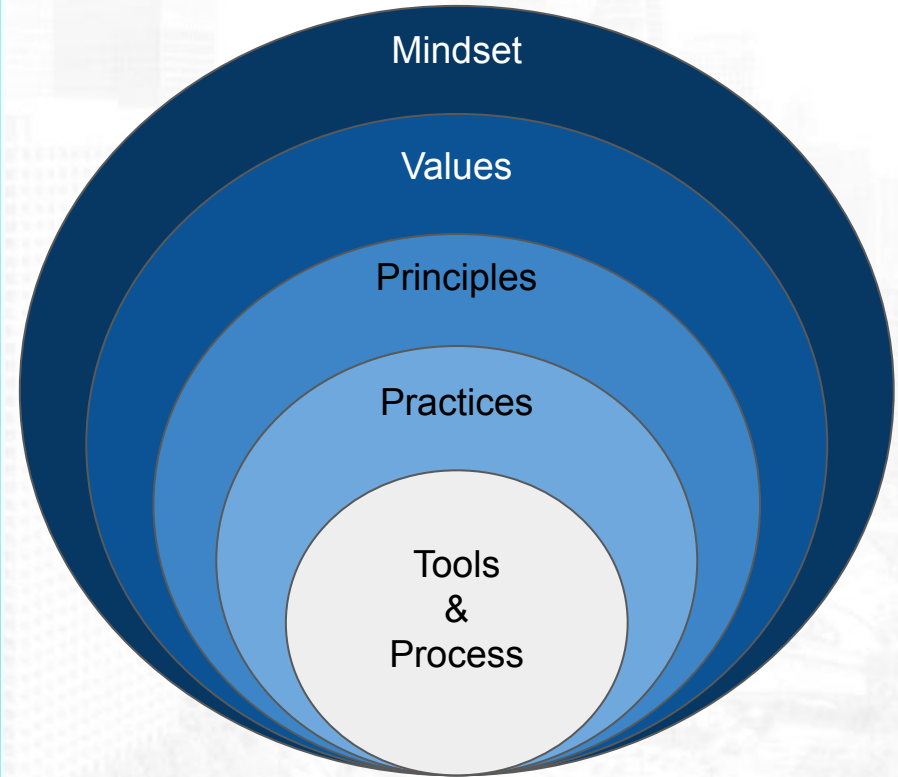
Problematic Background



Common problems of software development back then :

- Overcomplicated workflow and process
- Wasteful documents and meetings
- Unpredictable business demand
- Rigid development process
- Resulting in a manifesto with 4 values and 12 principles.

Agile in a nutshell



Further Readings

- <https://agilemanifesto.org/history.html>
- <https://www.agilealliance.org/agile101/the-agile-manifesto/>

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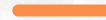
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the Agile Manifesto



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4 Agile Values



4

12 Agile Principles

4 Agile Values

Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Individual and Interaction over Processes and Tools

- Value or waste emerge as the result of individual action within the team and their interactions
- Established processes and acquired tools serves as the impact multiplier



Working Software over Comprehensive Documentations



- The most important thing is what value we brings to the customer
- Best way to measure value is to put our product in customer's hand



Customer Collaboration over Contract Negotiation



Healthy relationship between
developers and customer helps
promote productive feedback
Frequent collaboration helps
narrowing the gap between
development result and expectation



Responding to Change over Following a Plan

- What make sense back then might not so relevant now, remember VUCA
- Embrace unexpected changes and adjust plan accordingly



Further Readings

- <https://agilemanifesto.org>
- <https://www.agilealliance.org>
- <https://www.scrum.org/resources/blog/how-live-4-values-agile-manifesto>

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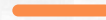
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12 Agile Principles

Principles behind the Agile Manifesto

We follow these principles:

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

Business people and developers must work together daily throughout the project.

Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

Working software is the primary measure of progress.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Continuous attention to technical excellence and good design enhances agility.

Simplicity--the art of maximizing the amount of work not done--is essential.

The best architectures, requirements, and designs emerge from self-organizing teams.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Customer Centric

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

- Customer first mindset, avoid taking assumption
- Break down work into smallest valuable piece of functionality and deliver value continuously

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter

- Deliver small increment frequently
- Establish shorter feedback cycle
- Fail fast and cheap

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

- Validate hypothesis, changes plan accordingly
- Focus on maximizing the value for the customer



Promote Collaboration

Business people and developers must work together daily throughout the project.

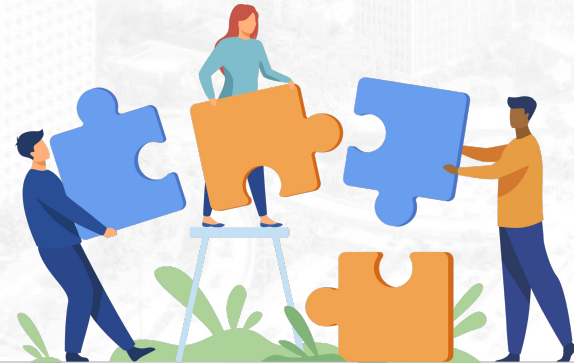
- Cross-functional team allows development flexibility and independence
- Team alignment is the key to achieve effective product development and faster problem discovery

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

- Nonverbal communication transfer more information compared to verbal communication
- In-person communication help increases trust and collaboration among team

Build projects around motivated individuals. Give them the environment and support they need and trust them to get the job done.

- Organization support help maximize team's productivity
- Safe environment promote innovation enabling effective and efficient development



Deliver Value

Working software is the primary measure of progress.

- Measure on outcomes not output, the value not the volume nor velocity
- While other metric might be helpful for the team internal improvement

Continuous attention to technical excellence and good design enhances agility.

- Agile focus on both functionality and usability
- Better quality prevent rework allowing the developers spent more effort on value generating activities



Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

- We prefer steady and sustainable development rather than fast but brittle ones
- All stakeholder should have the same vision towards the product goal

Simplicity the art of maximizing the amount of work not done is essential.

- Reduce wasteful development, processes, events, meetings, documentations as much as possible
- Do the easiest, simplest, the most straightforward approach as possible

Unleash Team's Maximum Potential

The best architectures, requirements, and designs emerge from self-organizing teams

- Agile empower people by promoting self-organized team
- Important decision should be made by the person doing it

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

- Periodically do reflection to discover any chances of improvement
- Committed to take improvement action even small ones but continuously



Further Readings

- <https://agilemanifesto.org>
- <https://www.agilealliance.org>

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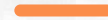
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Topic 2

Cynefin Framework and Stacey Matrix

Outline

1

Stacey Matrix



2

Cynefin Framework



3

Scrum Intro

Stacey Matrix*

**“If you only have a HAMMER,
you tend to see every
problem as a NAIL”
- Abraham Maslow**

Understanding our own Context



1 It would be great if you have read all the references

2 Prepare Stationary
(Pen ,Paper, Ruler, Etc)

3 Imagine 1 Situation that we want to asses
(Product, Project, Team, company, Etc.)

Based on our own Case

Steps to do:

Imagine 1 of your case

Find Gap of these 3 variables:

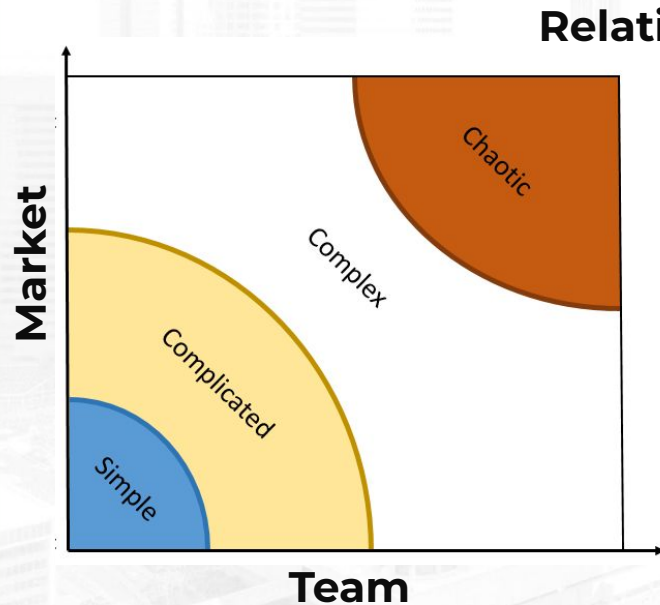
- Variabel X: Skill of your Team = 0-10
- Variabel Y: Market Landscape = 0-10
- Variabel Z: People Relationship = 0-10

Drawing your Vector Point in the Matrix

A faded, grayscale background image of a dense city skyline with numerous skyscrapers.

Let's Exercise!

Modified Stacey Matrix*



Due to VUCA,
you will keep flowing &
switching between the
domains.

**disclaimer: this exercise base on modified version of stacey matrix (5 areas simplified to 4)*

Outline



1

Stacey Matrix

2

Cynefin Framework

3

Scrum Intro

Cynefin Framework



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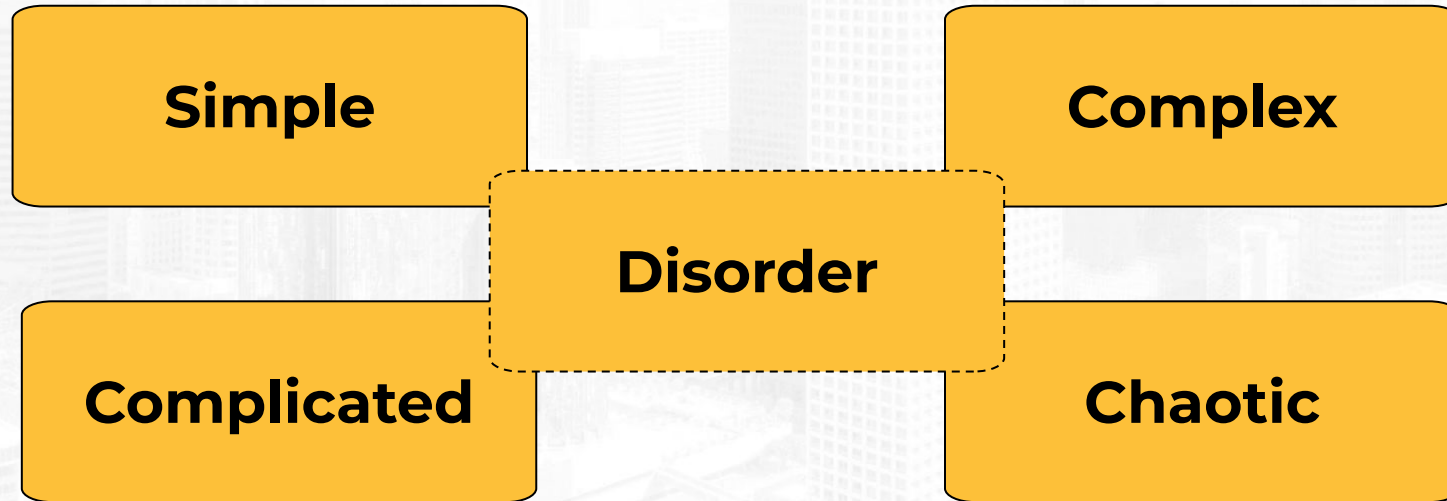
Decision Making And Problem Solving

A Leader's Framework for Decision Making

by David J. Snowden and Mary E. Boone

<https://hbr.org/2007/11/a-leaders-framework-for-decision-making>

Cynefin Framework



Let's Exercise!



If its broken, what kind of situation it will be?

Let's Exercise!



Is that Complex? Or Chaotic?

Cynefin Framework

Simple: Best Practice

you have capability to solve it
Don't oversimplified
Known-knowns zone

Complicated: Expert Practice

Known-unknowns zone
Multi Opinions & Answers
Need an area expert

DISORDER

Complex: Emergence Practice

Constant unpredictable flux
Unknown-unknowns
problems keep arrive
continuously monitor, aware & adapt

Chaos: Novel Practice

High turbulence & tension with no time
Immediate action + leadership test
clear, direct communication
we can't save everybody, unless we are superhero

Outline



Stacey Matrix



Cynefin Framework



Scrum Intro

Why Scrum.. ?

Scrum Framework

Complex: Emergence Practice

Constant unpredictable flux

Unknown-unknowns

problems keep arrive

continuously monitor, aware & adapt

Scrum Framework

A lightweight framework that helps People, generate adaptive solution for complex problems

3 Pillars of empiricism:

1. Transparency
2. Inspection
3. Adaptation

In a Summary

After understanding VUCA, we need to understand what is complex domain & what makes it different than the others (simple, complicated, chaos).

Awareness about complex problems, will give us a clear direction what mechanism & workflow that more suitable in that domain.

Scrum as a Lightweight framework that dedicated for complex problem. It can help us find adaptive solutions with transparency, inspection & adaptation.



References

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<https://hbr.org/1986/01/the-new-new-product-development-game>
https://adaptknowledge.com/wp-content/uploads/rapidintake/PI_CL/media/Stacey_Matrix.pdf
<https://scrumguides.org/scrum-guide.html>
<https://medium.com/enthusiasticscrum/scrum-isnt-going-to-fix-your-problems-15c10893d447>
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<https://www.mathsisfun.com/algebra/functions-maxima-minima.html>
<https://www.infoq.com/news/2019/07/refactoring-organizations/>

Professional = Theory + Practice

Outline

1

Stacey Matrix



2

Cynefin Framework



3

Scrum Intro

Topic 3

Scrum in a Nutshell

Outline

1

Empiricism
Thinking



2

Lean
Thinking



3

Scrum
Values



4

Scrum
Framework

Empiricism Thinking

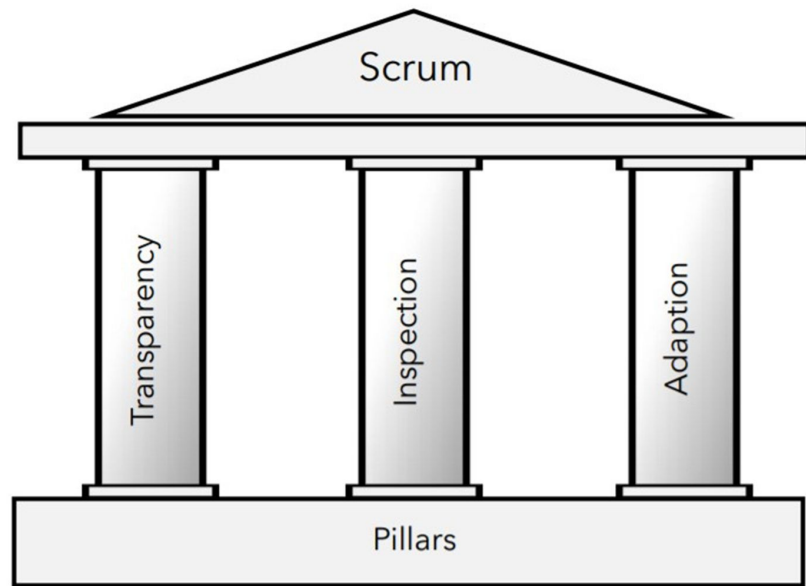
Scrum

The Base



Scrum is founded on **empiricism** and **lean thinking**.

Empiricism asserts that knowledge comes from experience and making decisions based on what is observed. Lean thinking reduces waste and focuses on the essentials.



Empiricism

empiricism

/ɛm'pɪrɪsɪz(ə)m/ 

noun

1. the theory that all knowledge is based on experience derived from the senses. Stimulated by the rise of experimental science, it developed in the 17th and 18th centuries, expounded in particular by John Locke, George Berkeley, and David Hume.

Powered by [Oxford Dictionaries](#)

3 Pillars of Empiricism

TRANSPARENCY

having access to knowledge and
a shared understanding

INSPECTION

trying something out based on that
knowledge

ADAPTATION

trying something else based on the
results



Transparency

Transparency, as used in science, engineering, business, the humanities and in other social contexts, is operating in such a way that it is easy for others to see what actions are performed. [Transparency implies openness, communication, and accountability.](#)
(wikipedia)

Why it is important

Transparency empowers, builds trust, and supports management decisions and drives factor behind this behaviour is the agile values of honesty, trust, integrity and respect.

Examples

- Presenting the facts as is
- No hidden agenda
- Being honest

Inspection

An **inspection** is, most generally, an organized [examination](#) or formal evaluation exercise.
(*wikipedia*)

Why it is important

The needs to evaluate to detect undesirable variances. The inspection can be done for the product, processes, people aspects, practices, and continuous improvements.

Examples

- shows the product to the customer in order to gather valuable feedback

Adaptation

Adaptation is the physical or behavioural characteristic of an organism that helps an organism to survive better in the surrounding environment.

Why it is important

In Agile world, we always embrace and Adapt changes, so that we can constantly improve. Adaptation means that we change what does not work or what could work better

Examples

- Adjustment the way we work during COVID19
- Changes in product scope following the user feedback

Transparency enables inspection.
Inspection without transparency is
misleading and wasteful. Inspection with
Adaptation is Pointless.

- Without **Transparency**, the data may not reflect the true reality and what people inspect will be incorrect and what they adapt to will be wasteful.
- Without **Inspection**, the decisions to adapt may not be supported by hard evidence and real change becomes more difficult.
- Without **Adaptation**, it will get frustrating and no changes happen.

Outline

1

**Empiricism
Thinking**



2

**Lean
Thinking**



3

**Scrum
Values**



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**Scrum
Framework**

Lean Thinking

Lean Thinking



Lean thinking is a way of thinking about an activity and seeing the waste inadvertently generated by the way the process is organized.

It is a strategic approach that focuses on dramatically improving flow in the value stream and eliminating waste.

Businesses in all industries and services, including healthcare and governments, are using lean principles as the way they think and do.



Outline

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**Empiricism
Thinking**



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**Lean
Thinking**



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**Scrum
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Framework**

Scrum Value

Scrum In a Nutshell



Scrum is a **lightweight framework** that helps people, teams and organizations generate value through adaptive solutions for complex problems.

It originates from the moment in a *rugby* game where suppose during any violation, or if the ball goes out of the ground, the team players lock their arms, heads down aim at taking possession of the ball.



Scrum

The Values

Successful use of Scrum depends on people becoming more proficient in living five values. These values give direction to the Scrum Team with regard to their work, actions, and behavior.



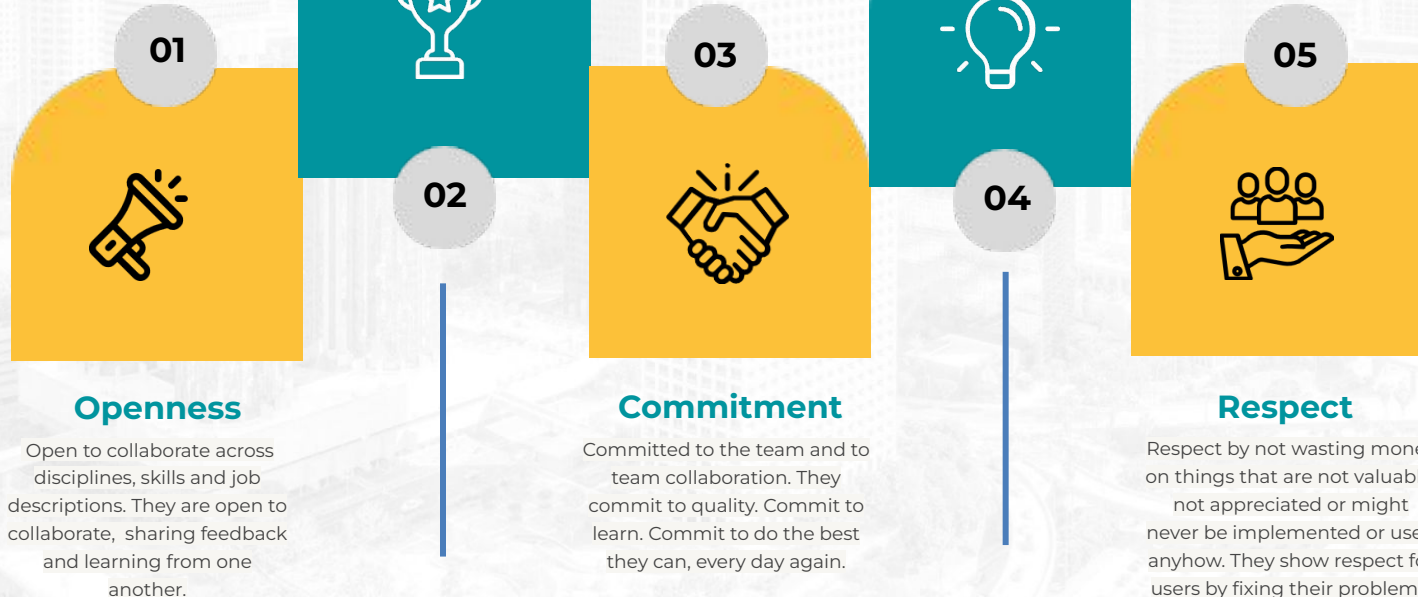
© Scrum.org

Focus

Focus on the work needed to get things done. They focus on the simplest thing that might possibly work.

Courage

Courage in admitting that nobody is perfect. Courage to change direction. Courage to share risks and benefits



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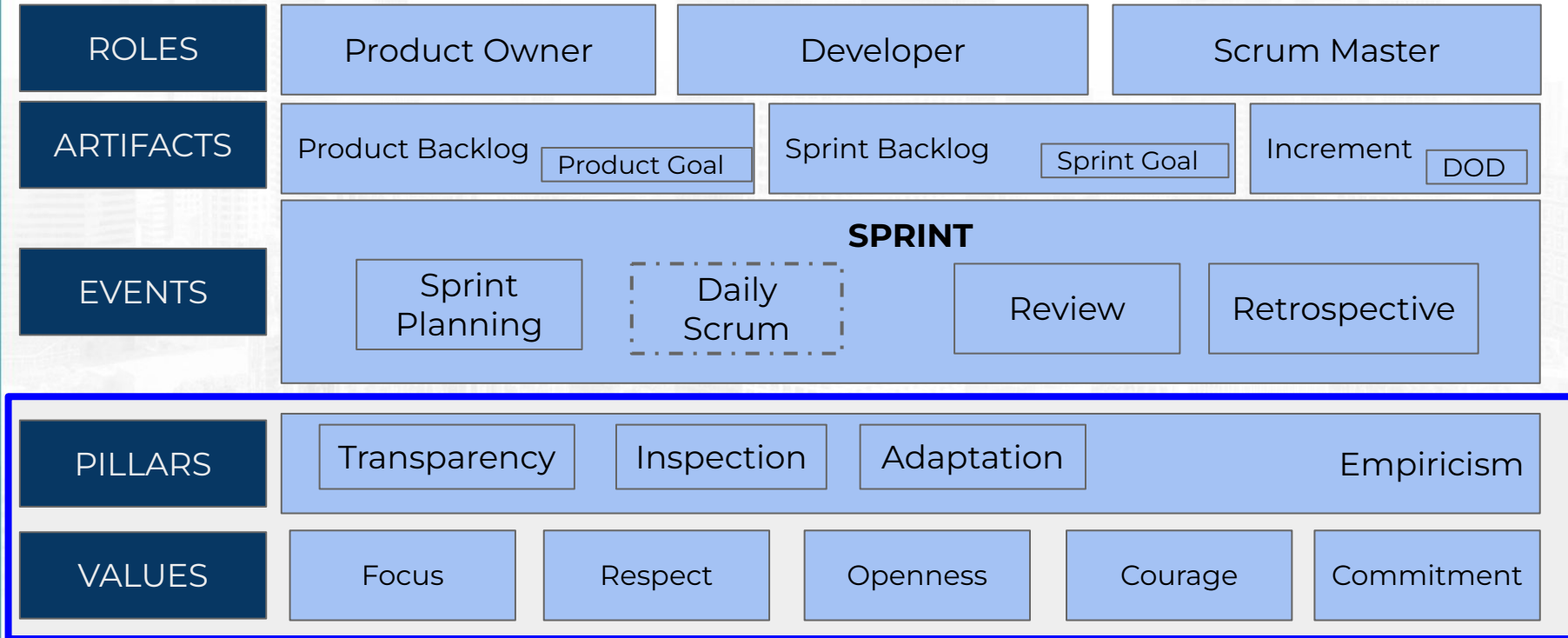


4

**Scrum
Framework**

Scrum Framework

Scrum Framework



Framework

framework

/ˈfreɪmwɜːk/ 

noun

1. an essential supporting structure of a building, vehicle, or object:
"a conservatory in a delicate framework of iron"

Similar

frame

substructure

structure

skeleton



Powered by Oxford Dictionaries

Scrum

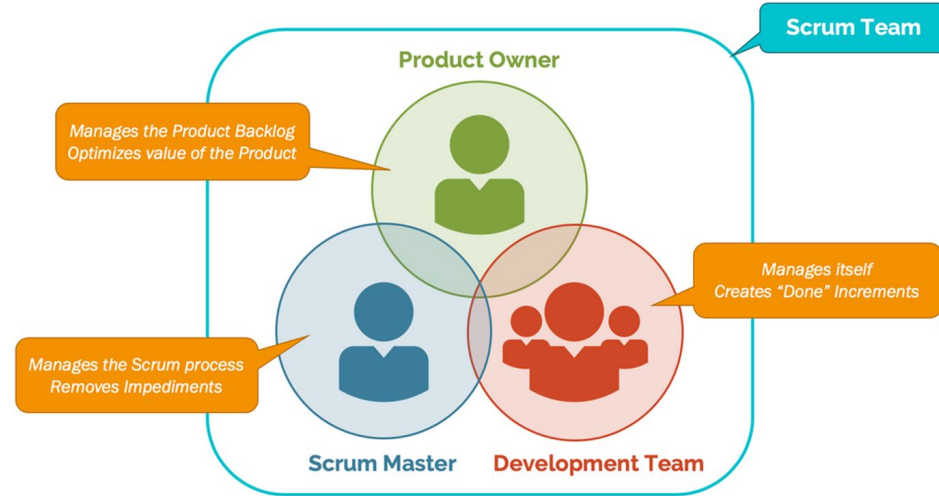
The 3 Accountabilities



The fundamental unit of Scrum is a small team of people, a Scrum Team.

As a Scrum Team :

- Focused on one objective at a time, the Product Goal
- Cross-functional, meaning the members have all the skills necessary to create value each Sprint
- Self-managing, meaning they internally decide who does what, when, and how
- Typically 10 or fewer people
- Accountable for creating a valuable, useful Increment every Sprint



Scrum

The Events



Optimally, all events are held at the same time and place to reduce complexity

Each event in Scrum is a formal opportunity to inspect and adapt Scrum artifacts. Events are used in Scrum to create regularity and to minimize the need for meetings not defined in Scrum.

Scrum Events Timebox

	Sprint Planning		2h	4h	6h	8h
	Sprint Review		1h	2h	3h	4h
	Sprint Retrospective		45min	90min	135min	3h
	Daily Scrum		15min	15min	15min	15min
	Sprint Length		1 Week	2 Week	3 Week	4 Week

Scrum

The Artifacts

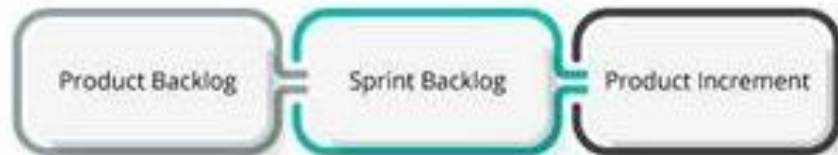


Scrum's artifacts represent work or value

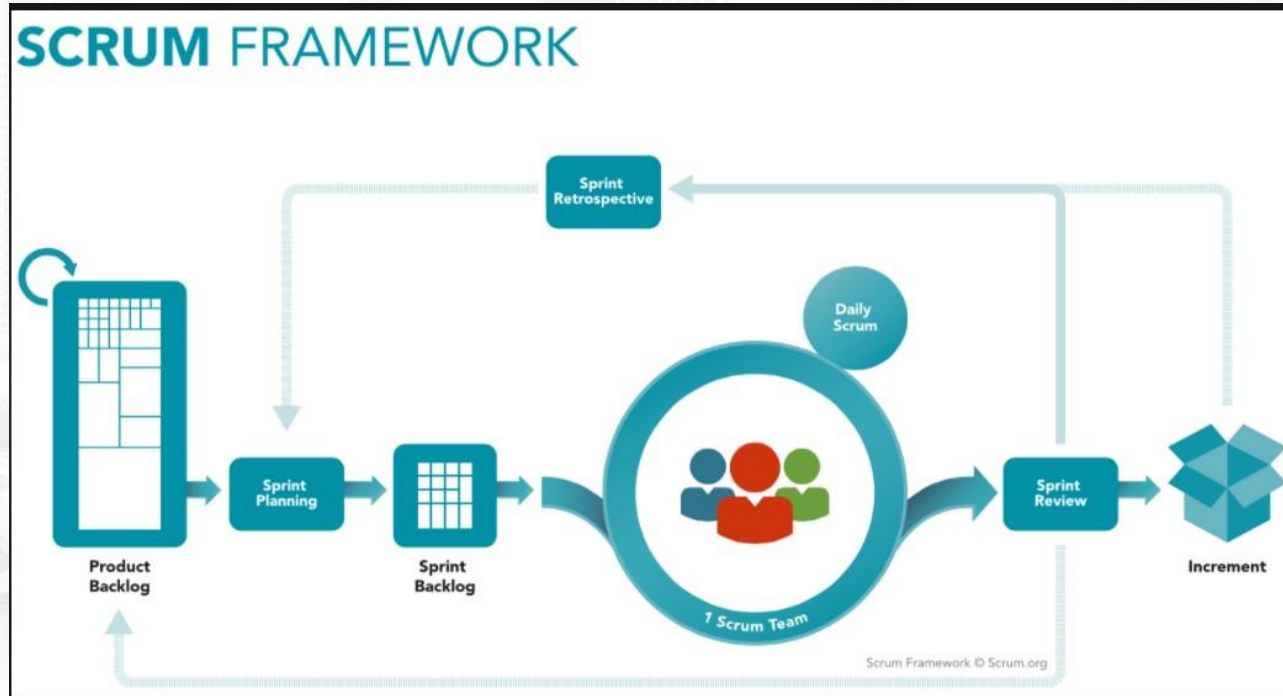
Each artifact contains a commitment to ensure it provides information that enhances transparency and focus against which progress can be measured:

- For the Product Backlog it is the Product Goal
- For the Sprint Backlog it is the Sprint Goal
- For the Increment it is the Definition of Done.

Scrum Artifacts



How it works



How it works

Adaptation is the goal



Event	Inspection	Adaptation
Sprint Planning	Product Backlog (Past performance) (Availabilities) (Retrospective Commitments) (Definition of Done)	- Forecast - Sprint Backlog - Sprint Goal
Daily Scrum	Sprint Progress (toward the Sprint Goal)	- Daily Plan - Sprint Backlog
Sprint Review	- Product Increment - Product Backlog (& progress) - Market & business conditions	Product Backlog
Sprint Retrospective	- Team & collaboration - Technology & development - Definition of Done	Actionable improvements (next Sprint)

(Gunther Verheyen – Ullizee-Inc)

References

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- <https://guntherverheyen.com/the-scrum-values/>
- <https://www.agilealliance.org>
- <https://www.toolsqa.com/agile/scrum-framework/>

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Thank You